

Udemy ML, DL, NLP Course

1

Sec - 39

PCA (Principal Component
Analysis)

Lec 1 - Curse of
Dimensionality

Lec 2: - Feature selection
Extraction

PCA (Dimensionality
reduction)

① Curse of Dimensionality
Dataset: - 500 features

M_1

M_2

M_3

M_4

M_5

M_6

=> Price of the house

2

House size

no. of bedrooms

no. of bathrooms

3

3 features

6 features

9 features

15 features

M_1

Acc 1

M_2

Acc 2

M_3

Acc 3

M_4

Acc 4

M_5

50 features

M_0

Accuracy of 1

=> Model performance

Loss

Loss

two different ways to (3)
solve curse of dimensionality:

- (1) Feature selection
 - (2) Dimensionality Reduction (PCA)
- Imp Features
-

⇒ Feature selection vs
Feature

→ Dimensionality
reduction

- (1) prevent → curse of dimensionality
- (2) Improve the performance of model
- (3) Visualize the data → understand the data

3d

1 sub

100d



3d

2d

=> feature selection

X

Y

—

—

—

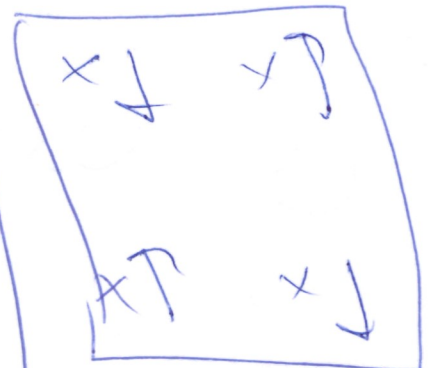
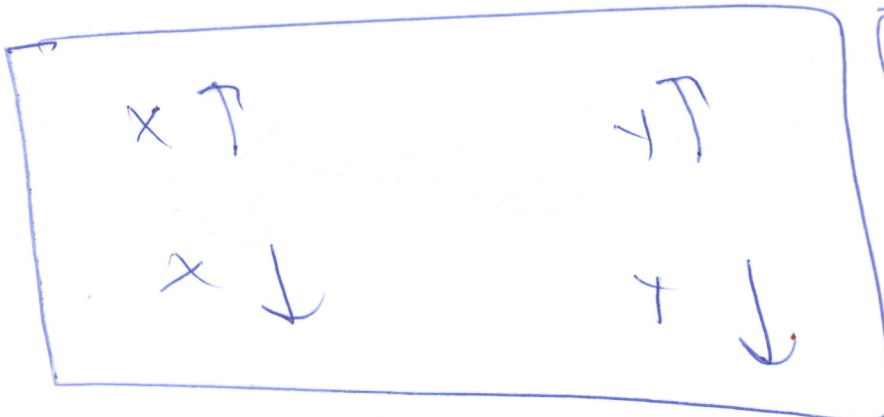
—

—

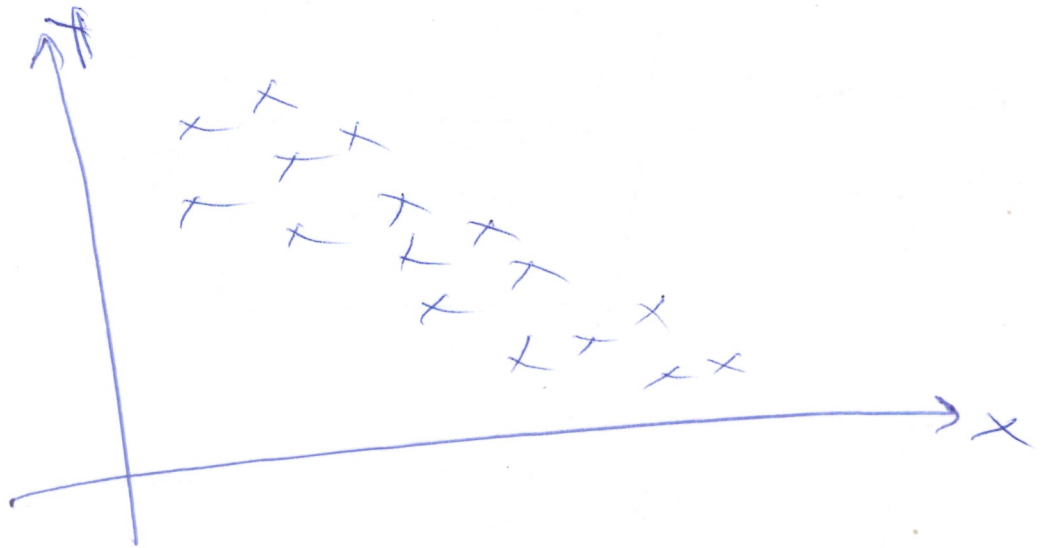
—

—

—



(9)



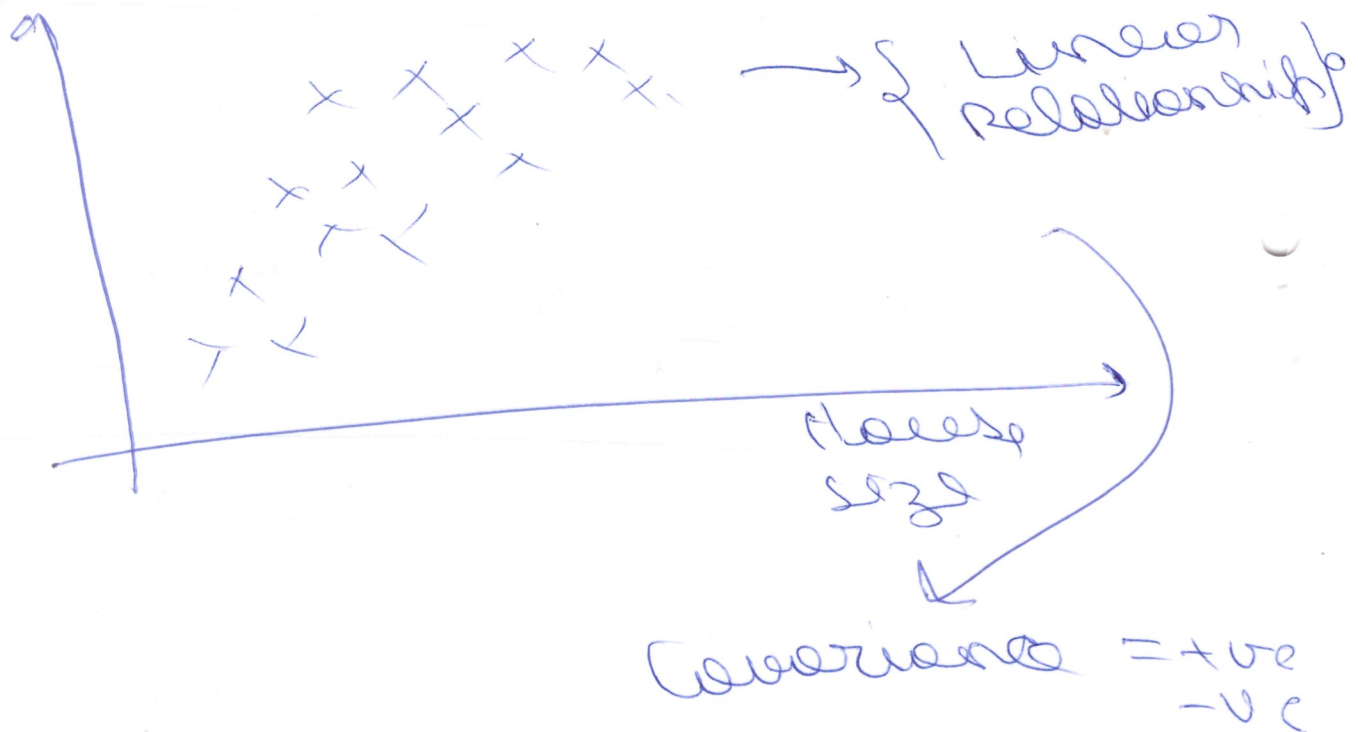
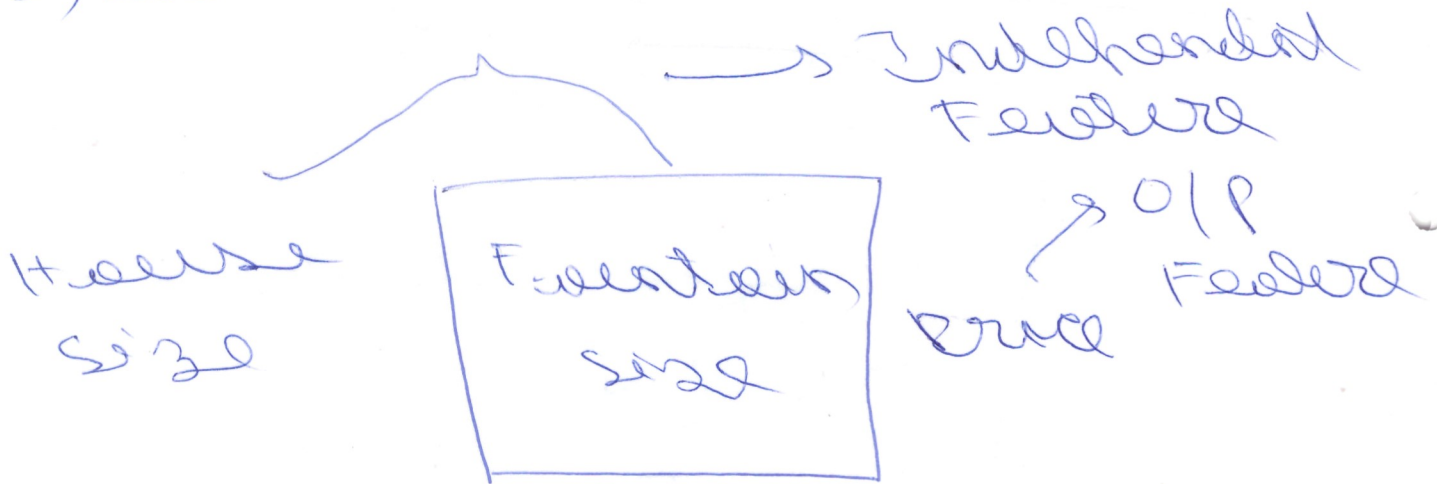
$$\text{Cov}(x, y) = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{n - 1} = +ve$$

 ≈ 0

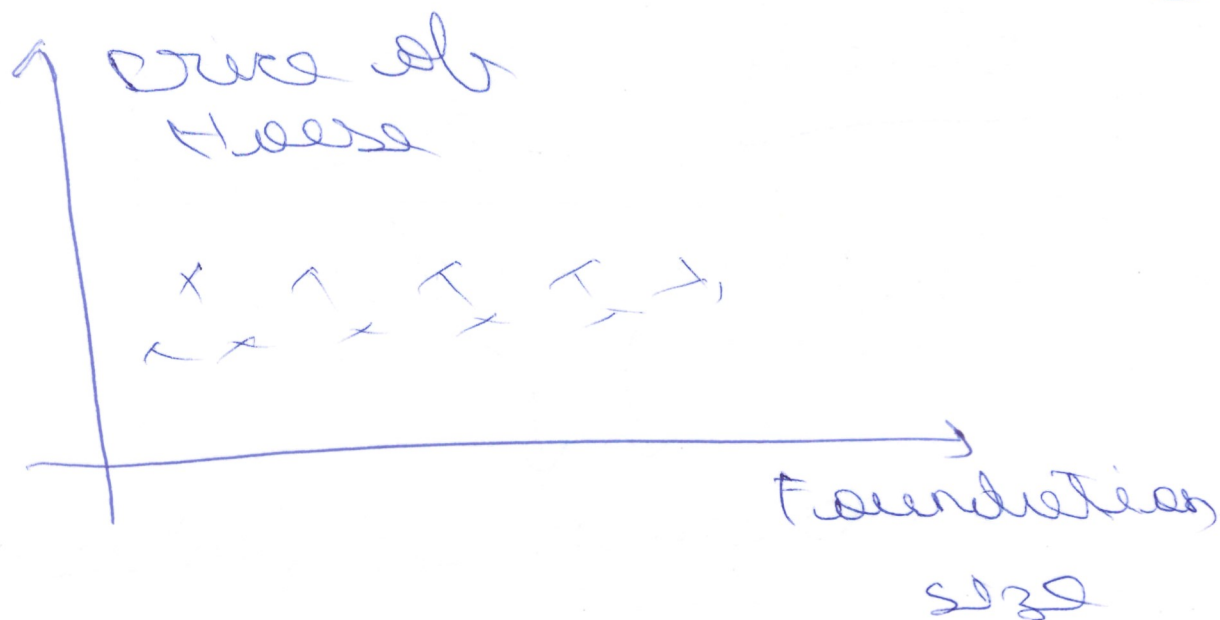
Pearson correlation coefficient = $\frac{\text{Cov}(x, y)}{\sigma_x \cdot \sigma_y} = -1 \text{ to } 1$

Use more features the
 will $Q+1$ the more
 correlated x y is. ⑥

⇒ Reduced Housing



2



⇒ Feature Extraction

Room size	No. of rooms	Price



Transformation
to extract new features

cruc

cruc

Domain Expertise